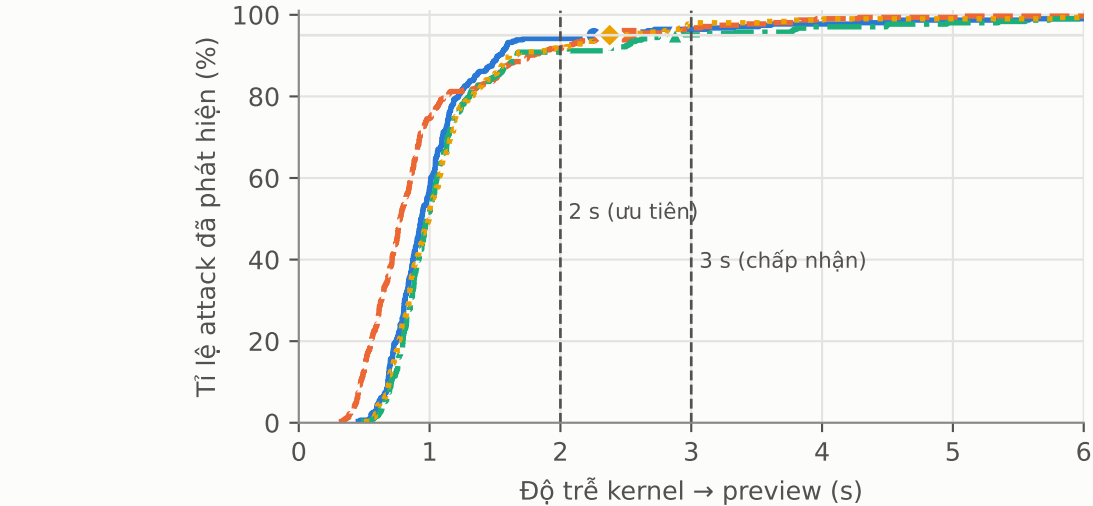
Tỉ lệ attack đã phát hiện (%)
100
80
60
40
20
0
0
1
2
3
4
5
6
2 s (ưu tiên)
3 s (chấp nhận)
Độ trễ kernel → preview (s)
pulse-stable-refreshed (k2of2, α×10 + calib. mới): p95 2.26 s, n = 310
pulse-stable-confirm (CUSUM b3/h6 cap, 6–24/s): p95 2.53 s, n = 314
pulse-peer-discount-confirm (k2of2 α×5 + replica 30 s, 6–24/s): p95 2.88 s, n = 306
pulse-final-system-confirm (k2of2 α×5 + replica 30 s, ML của hệ thống cuối, 6–24/s): p95 2.38 s, n = 308